Sprint 1 – Smart Home Media Server Infrastructure

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Sprint 1 Goals

- Establish core server infrastructure
  - Deploy Docker environment
  - Configure reverse proxy (Nginx Proxy Manager)
  - Implement Cloudflare Tunnel for secure external access
  - Deploy and test Nextcloud and Plex services

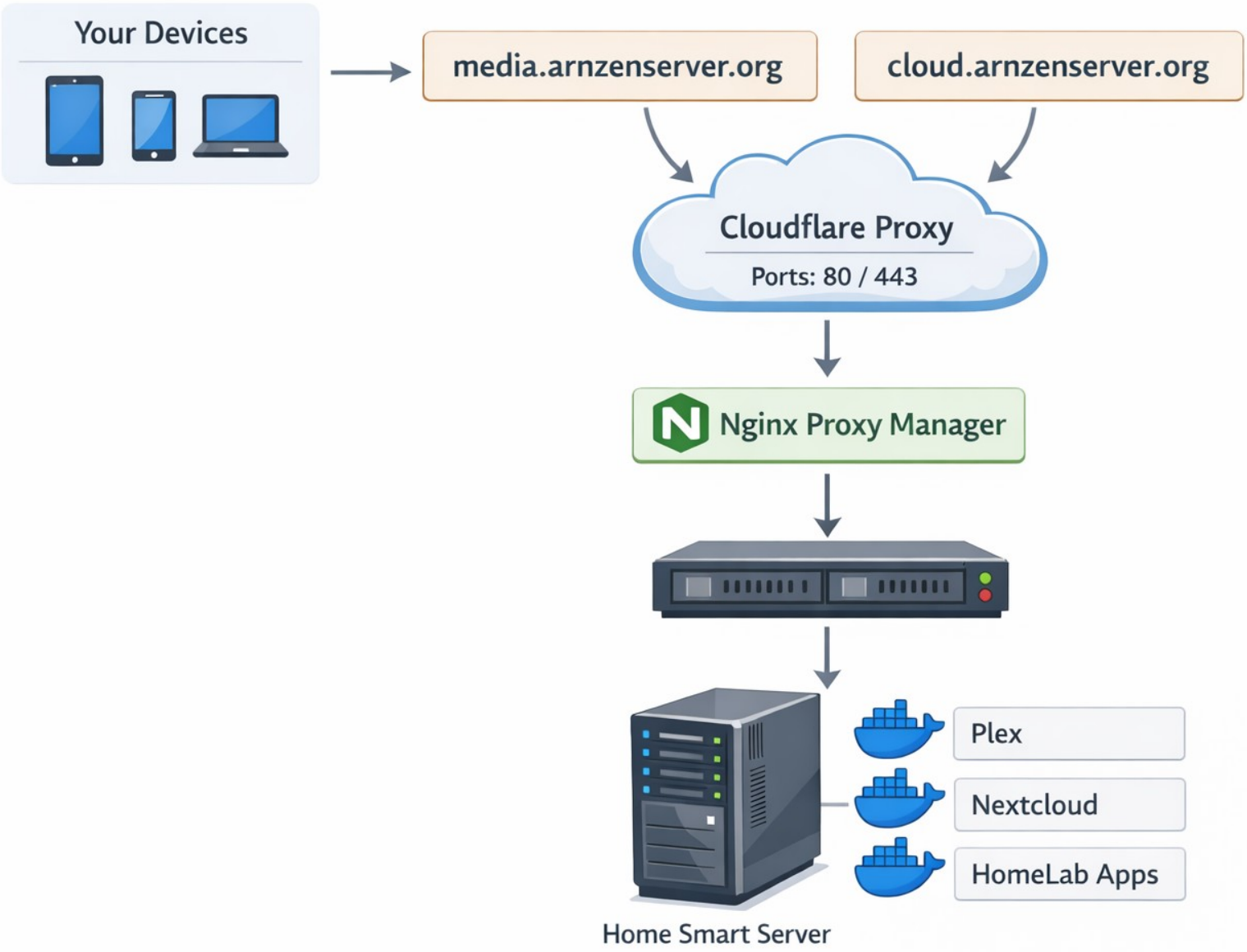
Setup

- Ubuntu Server running on local machine
- Docker installed for containerized services
- Nginx Proxy Manager for handling requests
- Cloudflare Tunneling to not expose ports
- arnzenserver.org domain for access

Architecture

Devices Access: cloud.arnzenserver.org, media.arnzenserver.org

- Requests → Cloudflare (DNS + Proxy) → encrypted Cloudflare Tunnel → NPM → Home server.
- Traffic arrives at Nginx Proxy Manager (NPM).
- NPM forwards requests internally (HTTP over Docker bridge network)


Your Devices
media.arnzenserver.org
cloud.arnzenserver.org
Cloudflare Proxy
Ports: 80 / 443
Nginx Proxy Manager
Plex
Nextcloud
HomeLab Apps
Home Smart Server

Docker

- Docker runs our applications in isolated containers:
- Nextcloud → internal container port 80
- Plex → internal container port 32400
- Traffic flows through Docker bridge network

Cloudflare

- Manages DNS and Proxies
- Cloud and Media subdomains are configured as proxied tunnel entries.
- This allows traffic requests without exposing my network

DNS management for **arnzenserver.org**

Review, add, and edit DNS records. Edits will go into effect once saved.

DNS Setup: Full ⓘ Import and Export Dashboard Display Settings

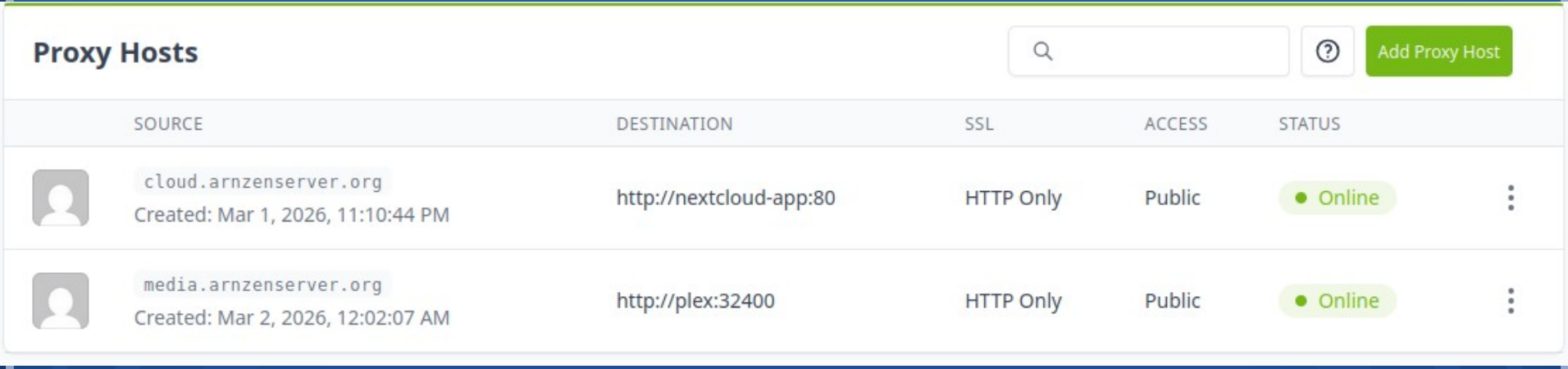
Search DNS Records

Add filter Search Add record

Type ⓘ	Name ⓘ	Content ⓘ	Proxy status ⓘ	TTL ⓘ	Actions
Tunnel	cloud	homelab	Proxied	Auto	Edit
Tunnel	media	homelab	Proxied	Auto	Edit

Nginx Proxy Manager (NPM)

- Running in Docker container
  - Listens on ports 80/443
  - Routes traffic based on hostname



Nextcloud

- Running in Docker container
  - Accessible internally via LAN
  - Accessible externally via cloud.arnzenserver.org
  - Handles Cloud Storage
  - Soon will be able to upload videos for plex

cloud.arnzenserver.org/apps/files/files

Search here ...

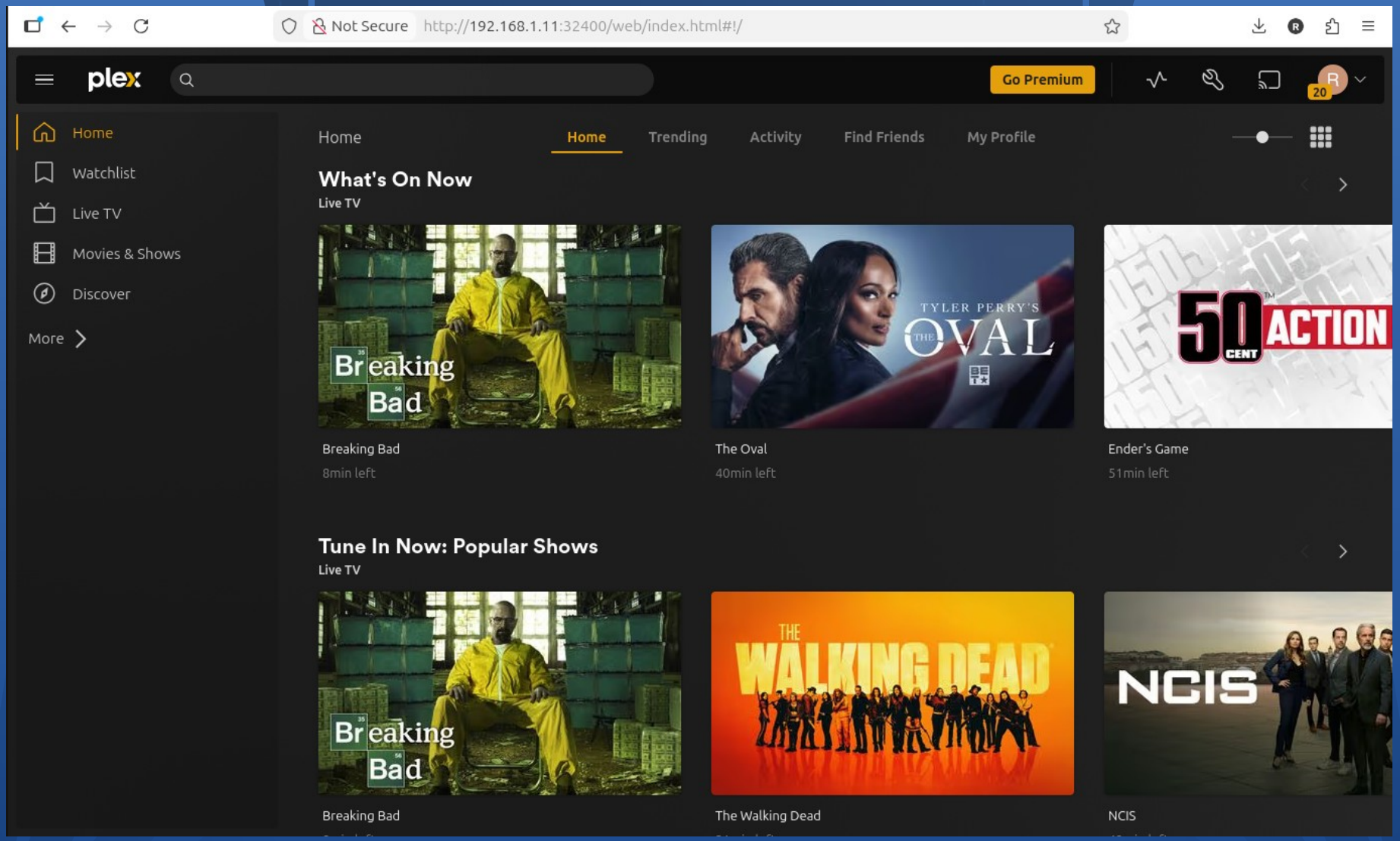
- All files
- Favorites
- Folder tree
- Personal files
- Recent
- Shares
- Tags
- Deleted files
- 58.1 MB used
- Files settings

+ New | All files | Type | Modified | People

Name	Size	Modified
Documents	1.1 MB	3 hours ago
Photos	5.4 MB	3 hours ago
Talk	0 KB	3 hours ago
Templates	22.4 MB	3 hours ago
Nextcloud.png	49 KB	3 hours ago
Nextcloud intro.mp4	3.8 MB	3 hours ago
Nextcloud Manual.pdf	24.5 MB	3 hours ago
Readme.md	< 1 KB	3 hours ago
Reasons to use Nextcloud.pdf	954 KB	3 hours ago
Templates credits.md	3 KB	3 hours ago
6 files · 4 folders	58.1 MB	

Plex

- Running in Docker container
  - Handles media streaming across devices
  - Accessible internally via 192.168.1.11:32400
  - Plex listens on port 32400 inside container
  - Runs on home network


Not Secure
http://192.168.1.11:32400/web/index.html#!/
plex
Go Premium
20
Home
Watchlist
Live TV
Movies & Shows
Discover
More
Home
Home
Trending
Activity
Find Friends
My Profile
What's On Now
Live TV
Breaking Bad
8min left
TYLER PERRY'S
THE OVAL
The Oval
40min left
50 CENT
ACTION
Ender's Game
51min left
Tune In Now: Popular Shows
Live TV
Breaking Bad
THE WALKING DEAD
The Walking Dead
NCIS
NCIS

Databases and Storage

- Nextcloud: Uses internal database container (MySQL)
  - Plex: Uses internal SQLite database (within its container)
  - Docker volumes are used for persistent storage seperate from container
  - Server file system stores media and cloud files

What is Working

- Docker is stable
  - NPM reverse proxy routing functional
  - Cloudflare Tunnel connected
  - Nextcloud accessible externally (HTTPS)
  - Plex accessible internally on LAN

Sprint 2 Focus

- Finalize Plex external HTTPS configuration
- Implement shared library for uploads
- Add monitoring (Prometheus/Grafana)
  - Structured data for server metrics
  - Will allow easy AI training
- Begin predictive maintenance metrics collection

Learning With AI

- I used AI to learn:
  - Cloudflare proxy and tunneling
  - Nginx reverse proxy
  - Container network bridge

Questions?